CAD ENGINEERING TECHNICAL INTERVIEW

EDA / CAD Engineering Projects

Logic Synthesis / Physical Design / Timing ECO / CAD Flow Automation

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Education & Research Interests

01 / PROFILE

Education

National Taiwan University
M.S. Candidate, Data Science Master Degree Program
Sep. 2025 – Expected Jun. 2027

National Chengchi University
B.S. in Computer Science
Double Major in Electronic Physics
Sep. 2021 – Jun. 2025

Research Interests

EDA / CAD & VLSI Design Automation

Logic Synthesis
Technology Mapping
Static Timing Analysis
Timing Closure
Physical Design
Incremental ECO
PPA

Algorithm & Software Development

C++ / Python / Tcl Flow Automation

Professional Software Engineering Experience

02 / WORK EXPERIENCE

Software development, simulation, and validation experience in R&D environments.

AIMobile Co., Ltd. (Inventec Group)

Apr. 2025 – Jul. 2025

Contract Software Engineer

Contributed to software development and system integration in an R&D environment.

Supported debugging and functional validation across software modules.

Neopower Technologies Co., Ltd.

Jun. 2023 – Mar. 2025

Contract Software Engineer

Developed a custom system simulator to reproduce operational and error scenarios.

Performed regression checks, log analysis, and customer-side A/B validation.

Selected Projects & Discussion Topics

03 / PROJECT OVERVIEW

01 Master's Research — Logic Synthesis (Front End)

Joint Single- and Multi-Output Cell Mapping via Differentiable Optimization

EDA Tool Development / Boolean Matching / Area–Delay Optimization

02 ICCAD 2025 Problem C — CAD Methodology (Back End)

STA-Driven Incremental Timing ECO Search

OpenROAD / OpenSTA / Python–Tcl Flow Automation / Heuristic PPA Search

Joint SO / MO Mapping: Research Pipeline

01 / RESEARCH SCOPE

Discover FA/HA opportunities, pre-select non-overlapping MO candidates, then let the optimizer choose SO or BIND.

Problem

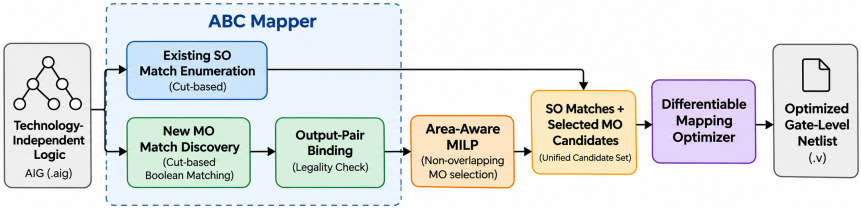
Single-output cut retention does not target two-output FA/HA opportunities.

ABC-Based MO Discovery

A new MO candidate search inside the ABC Mapper. Existing SO enumeration stays unchanged.

MILP Pre-selection

Keep a non-overlapping, area-aware subset of legal MO candidates. SO matches bypass this step.



Why an Independent MO Discovery Path?

03 / MOTIVATION

SO-oriented cut pruning may remove cuts useful for joint FA/HA output matching.

ABC Mapper SO Ranking

Cuts are retained for single-output area and delay, not for complementary FA/HA output pairs.

ABC-Based MO Discovery

Keep additional MO cuts and CON/SN truth projections without changing the ABC Mapper's SO path.

FIGURE SPACE

SO-pruned cut vs MO-retained cut

Two output roots with a shared legal 3-leaf input support.

MO discovery is not equivalent to scanning the final selected SO cover.

MO Candidate Discovery in ABC

04 / CORE ALGORITHM

Identify pairs of logic outputs that can be implemented by a single multi-output standard cell.

01 / MO Cut Discovery

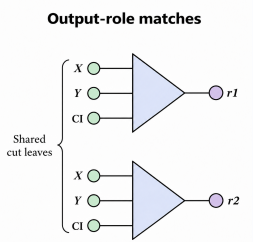
Search for FA/HA-compatible logic cuts independently of SO-oriented pruning.

02 / Output Pair Matching

Find SUM and CARRY functions with shared inputs and compatible pin mapping.

03 / Candidate Validation

Verify functional and structural legality, then identify the SO logic that the MO cell could replace.



Two output roots share the same input boundary (X, Y, CI) and match the SUM/CARRY functions of a full-adder cell.

Baseline: File-Order Candidate Loss

File-order greedy filtering ignores estimated area benefits of conflicting MO candidates.

41,476
Original MO candidate BIND records

7,024
FIFO retained (offline simulation)

16.9%
Overall retention rate

Design	Original MO	FIFO Retained	Dropped	Retention
EPFL hyp	25,440	4,180	21,260	16.4%
EPFL square	3,658	544	3,114	14.9%
EPFL log2	2,798	400	2,398	14.3%
EPFL multiplier	2,190	349	1,841	15.9%
EPFL voter	632	106	526	16.8%
ISCAS c6288	258	37	221	14.3%
ISCAS c7552	102	11	91	10.8%
EPFL adder	6	1	5	16.7%
All 364 designs	**41,476**	**7,024**	**34,452**	**16.9%**

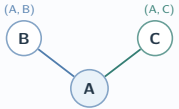
Source: MO-candidate census, 364 benchmark entries; simulated first-fit retention, not mapping-optimizer QoR.

Area-Aware MO Selection (MILP)

09 / GLOBAL CANDIDATE SELECTION

Maximize estimated area savings subject to non-overlapping MO candidates.

$$
\begin{aligned}
\max \quad & \sum_{i=1}^{N} \left(A_i^{SO} - A_i^{MO}\right) x_i \\
\text{s.t.} \quad & \sum_{i=1}^{N} a_{vi} x_i \le 1, \quad \forall v \in \mathcal{V} \\
& x_i \in \{0,1\}, \quad i = 1, \ldots, N
\end{aligned}
$$



$x_{AB} + x_{AC} \le 1$

Notation

x_i: select MO i; A_i^{SO}: SO witness area; A_i^{MO}: MO cell area;

$a_{vi} = 1$ iff AIG base node v belongs to MO candidate i; otherwise 0.

Implementation

Python + HiGHS (`highspy`); emit selected.matches.

Objective weights estimate local SO-to-MO benefits. Final QoR is measured after the mapping optimizer; MILP retention does not force every selected candidate into the netlist.

Current Mapping Flow & ADP Improvement

11 / CURRENT RESULTS

Current flow

MO candidate discovery in ABC

Boolean / structural legality

SO witnesses + area weights

MILP candidate pre-selection

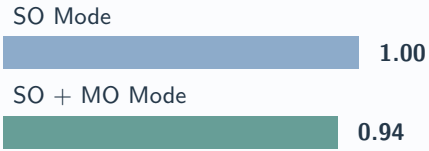
Mapping optimizer SO / BIND choice

Measure final mapped QoR after optimization; MILP weights are local estimates.

6%

Lower ADP vs. SO Mode

Normalized ADP (SO Mode = 1.00)



ADP = area × delay. Comparison is SO mode versus SO+MO mode.

PROJECT 02 / ICCAD 2025 CAD CONTEST / PROBLEM C

Incremental Placement Optimization Beyond Detailed Placement

Simultaneous Gate Sizing, Buffering, and Cell Relocation

MY PROPOSED APPROACH

STA-Driven Incremental Timing ECO Search

Use OpenROAD / OpenSTA with policy-controlled ECO moves and global PPA evaluation.

Multi-Objective Timing ECO Optimization

02 / PROBLEM DEFINITION

Improve setup TNS toward zero while controlling total power, HPWL, and cell displacement.

OPTIMIZATION OBJECTIVES

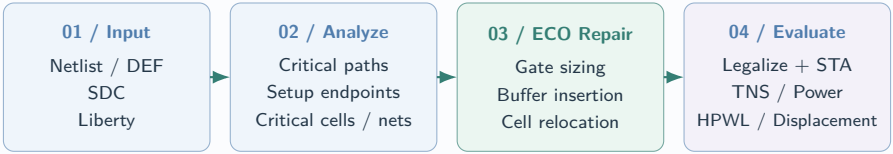
01 / Setup TNS
Reduce negative setup slack at violating endpoints.

02 / Total Power
Avoid excess power from ECO changes.

03 / HPWL
Control estimated wirelength after ECO.

04 / Displacement
Limit cell movement and keep placement legal.

BASIC ECO EVALUATION PIPELINE



Key challenge: a timing fix can worsen power or physical cost.

Five Automatic ECO Strategy Candidates

02 / CANDIDATE SET

The candidate is an automatic repair policy and editable region, not a manually chosen cell replacement.

S1 / Global Setup

Whole design; automatic upsizing and VT selection.

S2 / Path-Restricted

Top-K path cells + nets; localized setup repair.

S3 / Overlap-Aware

Top-I shared cells, nets fixed; sizing / VT repair.

S4 / Local Buffering

Top-K nets, original cells fixed; buffer insertion and load splitting.

S5 / Power Recovery

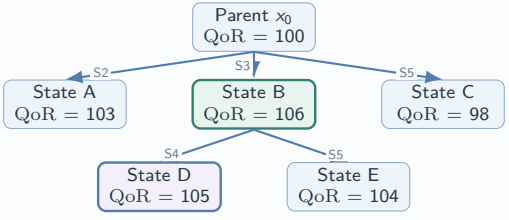
Slack-rich cells / wider region; lower-power equivalent cells.

STA-Guided ECO Search: Best-First Branch List

02 / HEURISTIC SEARCH POLICY

No better child this round? Continue from the best unexplored state in the Branch List.

Search Tree Example



Bounded Branch List: D:105 > E:104 > A:103

Global Best: B (106) | **Next:** D (105)

A lower-scoring intermediate state may lead to a better ECO sequence later.

Heuristic Search Steps

01 / Pop Parent

Highest-QoR unexplored state from the **Branch List**.

02 / Generate + Evaluate

STA-guided ECO moves, legalization, and global QoR score.

03 / Update States

Save improved **Global Best**; queue legal child states.

04 / Next Branch

Expand best branch, **even with no new Global Best**.

QoR Results & Conclusion

02 / EARLIER ECO EVALUATION

Setup-focused timing repair and the resulting power / wirelength trade-off.

Setup Timing

16.7%
Median Setup TNS Improvement

14.6%
Average Setup WNS Improvement

Power & Wirelength

8%
Total Power Reduction

+2%
HPWL Increase

Conclusion

ECO operators can improve setup timing and power while accepting a small wirelength trade-off.

THANK YOU

Questions & Discussion